Not at a distance, but nowhere: non-locality beyond entanglement, in twelve phenomena

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Abstract

In physics «non-locality» mostly names one fact: the correlations of entanglement. In Mathematical Materiality (MM) it is the foundation: a non-local phase, not located in space and time and present everywhere, of which the local world is the count made from a point. Here «non-local» means not located in space and time; it does not mean action at a distance. This record collects, from MM's published records and the author's premises, twelve phenomena in which MM's reading uses the same non-local phase as the reason, and shows that entanglement is the case in which it rests most directly on a measured datum, not the only one. There are no new calculations: the result is the unity of the reading. The twelve phenomena:

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In brief

Mathematical Materiality (MM) starts from a non-local phase: reality as relations that are not in a place or at an instant, present everywhere. The local world is the same reality counted from a point. Entanglement is the case in which this phase rests most directly on a measured datum, but it is not the only one. This deposit collects twelve phenomena in which MM's reading uses the same non-local phase as the reason: the correlations of entanglement and their ceiling, the same laws everywhere, identical particles, the objects that two observers see alike, the boundary of the local, probability, the threshold beyond which an object stops being at one point, the present, space and time themselves, the limit of light, dark matter, and the boundary of a gravitating system. For each it says what is observed, how known physics reads it, what the non-local phase does, what rests on calculations already deposited and what remains open, and where the link comes from: a deposit, a premise of the author, or a proposal of the AI. There are no new calculations: the result is the unity of the reading.

The question

In physics the word «non-locality» mostly names one fact: the correlations of entanglement. Bell's theorem (1964), in the form also used in 2.68, shows that three assumptions together put a limit on the correlations between two distant systems: each system carries its own properties with it, no influence travels faster than light, and the choices of measurement are independent of the state of the source. Experiments exceed that limit, and physics treats non-locality as a property of entangled systems.

Elsewhere physics is local. Some very general facts it explains with separate assumptions: the homogeneity of space and time for the same laws everywhere, one field for each species of particle for identical electrons, the redundancy of copies in the environment for two observers who see the same object.

MM starts from the other end. Its foundation is the non-local phase; the local world is the same reality read from a point. If so, non-locality cannot appear in entanglement alone: it must be at work wherever something is in a place.

In the philosophical deposit 2.68 the declared role of the non-local phase was to give a reason for two facts: that the laws are the same everywhere, and that the two ends of an entangled system are one nexus. The author's premises and the other deposits, however, use it in many other places. This deposit puts them in a row. For each it separates what is a measured datum, what is a known theorem, what is a registered calculation and what is a premise, and it says whether the link with the non-local phase is already in a deposit, comes from a premise of the author, or is proposed here by the

AI.

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1. What «non-local» means in MM

In MM «non-local» means **not located in space and time**. It does not mean «at a distance», nor «faster than light». Space and time are not the container of the relation, but the way it is read.

The framework distinguishes a non-local phase and a local phase, but they are not two stages, one before and one after, nor two planes, one above the other. The non-local phase has no time, so it comes before nothing; it has no place, so it is not in a location from which a distance separates us. It is present in every thing, in every place and at every instant. The local phase is the same reality counted from a point. In earlier deposits the same thing is also called an «underlying level» or «step»; here, as on the entry page of the framework, it is said without the image of planes: present everywhere.

In the non-local phase there is only a logical distinction: space is not time. The two distinct terms have a single element in common, which MM calls their greatest common divisor (GCD), as 3 and 5 have only 1 in common. Between 3 and 5 there is no intermediary: nothing travels from one to the other, and the relation holds entirely and at once. When the relation is counted through its GCD, the multiples and the fractions form a matrix: this is the local world. Read by an observer inside it, the multiples appear as time and the fractions as space.

Two postulates of the framework, from its entry page (record 22667516), do the work in everything that follows:

- **Existing is reciprocal:** an entity exists for another only to the extent of what they have in common.
- **Copies have no distance:** three copies are the same unit counted three times, and between one copy and another there is no distance.

Hence the sentence that sums up the framework: **the local is the non-local counted**. Every distance is made of copies between which there is no distance.

The four senses of the word. In philosophy of physics «non-local» has at least four senses; 2.68 separates them as follows.

sense	what it means	in MM
non-location	the relation is not in a place or at an instant	the sense of MM
non-separability (Howard 1985; Teller 1986)	two systems have no separate states of their own	a consequence: two copies of the same unit are not two things
Bell non-locality (Bell 1964)	correlations that no account with separate properties, no influence faster than light and independent measurements reproduces	a datum; MM reads it with non-separability and with the loop (§3.1)
action at a distance	an influence going from here to there faster than light	excluded

What is not claimed. The non-local phase transmits no signals and no influences. The events, settings and records of experiments are in spacetime as always (2.48, §3).

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2. One mechanism, seen in three ways

Explanatory premise (AI). The organisation in three ways is the AI's; the premises are the author's. The author accepted it on 28 October 2026.

In the twelve phenomena of §3 the non-local phase always works in the same way, but it is seen from three sides.

- **Way A, copies without distance.** What appears in many places is the same unit counted several times, and there is no distance between the copies. A likeness at a distance follows, which no message carried: the two ends of an entangled system, two places with the same laws, two observers who see the same chair, the same limit of light for everyone.

- **Way B, existence passes through a link.** An entity exists for another to the extent of what they have in common, and what they have in common is a non-local link. Where the link is missing there is no hidden object: for that observer there is nothing. For the same reason nothing can exist outside the web of relations, not even other universes. This is the hardest side of the framework, and it is at work in observed objects and in dark matter.
- **Way C, the reading is local and partial.** For the author it is the non-local basis that forces the present to a part of the matrix: the non-local link «switches on» only the part to which the observer belongs. The reading that follows is local and partial, and some of the limits we measure belong to it: the Tsirelson bound, which «is set by the local reading», the present, the threshold beyond which an object stops being at one point, and the boundary within which a system reads only its own field.

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3. The phenomena

Each subsection has six entries: what is observed; how known physics reads it; what the non-local phase does in MM's reading; where the link comes from; what rests on calculations already deposited; what remains open.

3.1 The correlations of entanglement and their ceiling (ways A and C)

- **What is observed.** Correlations beyond the Bell limit with the main loopholes closed: two electrons in laboratories 1.3 km apart (Hensen et al. 2015, $S = 2.42 \pm 0.20$) and atoms 398 m apart (Rosenfeld et al. 2017, $S = 2.221 \pm 0.033$). Beyond the limit also photons sent from a satellite to stations 1 203 km apart (Yin et al. 2017, $S = 2.37 \pm 0.09$) and tests with settings chosen by the light of stars and quasars (Handsteiner et al. 2017; Rauch et al. 2018), in these three cases assuming fair sampling of the detectors. The correlations cannot be used to send messages, and they do not exceed a second, higher limit: the Tsirelson bound, $2\sqrt{2}$ (Poh et al. 2015: 2.82759 ± 0.00051, with the central value 0.00084 below $2\sqrt{2} = 2.82843$). The 2022 Nobel Prize went to Aspect, Clauser and Zeilinger for experiments with entangled photons.
- **Known physics.** Quantum mechanics predicts them with the entangled state; the ceiling is Tsirelson's theorem (1980). Interpretations differ on what produces them.
- **What the non-local phase does.** The two ends are not two systems with properties of their own: they are two copies of the same nexus (way A). Of the three assumptions in the form used here, MM drops the first, separability, at the root: this is a premise of the framework, not a result of the theorem. The ceiling, instead, is set by the local reading of each end (way C).
- **Where the link comes from.** From the deposits: 2.48, 2.68 (§1 and §5), the overview (ch. 4).
- **What rests on calculations (MM 2.48 and 2.48.1).** If each end is read with the Born geometry, no correlation exceeds $2\sqrt{2}$ (a theorem, already in the literature). If the two ends carried only a common record written at departure, they would stay below 2, which experiments exceed. For the singlet, with coplanar projective measurements and the angular law assumed, the correlations are reproduced with every link inside light cones, through the common past of the two readings: nothing goes directly from one end to the other (the loop model; neighbours: Aharonov, Bergmann and Lebowitz 1964; Cramer 1986; Argaman 2010; Wharton and Argaman 2020). The price is Bell's third assumption: the choices of measurement are not independent of the state, because the future enters its preparation.
- **Open.** Non-separability alone does not give the numbers of the correlations: the Born reading at the two ends does. Experiments exclude the three assumptions together; they do not choose MM among the interpretations. Whether the model respects relativistic causality still has to be argued separately (2.48, §5). Whether the non-local phase and the loop are the same fact seen from two sides is a reading of the AI (2.68, §5).

3.2 The same laws everywhere and always (way A)

- **What is observed.** No measured difference between the laws in different places and epochs. A precise check, which concerns one constant, is the fine-structure constant: in the spectrum of a quasar at $z = 1.15$ it varies by no more than a few parts per million (Murphy et al. 2022: $\Delta\alpha/\alpha = 1.3 \pm 1.3 \pm 0.4 \times 10^{-6}$), and in today's atomic clocks by no more than about 10^{-18} per year (Lange et al. 2021).
- **Known physics.** The universality of laws is a starting assumption. Through Noether's theorem, the invariance of the laws under translations in time and space gives the conservation of energy and momentum; the theorem ties symmetries to conservation laws, it does not explain why the symmetries hold. Physics assumes universality and checks it, without giving it a reason.
- **What the non-local phase does.** Every «here» and every «now» is a copy of the same nexus, and between two copies of the same unit there is no distance. Two places cannot have different laws for the same reason that two copies of 1 are not two different units. **The universality of laws is non-locality seen from inside the local.** Laws need no guardian: 3 and 5 do not «obey» their GCD, they are that fact.
- **Where the link comes from.** From the deposits: 2.68 and 2.68.1, §7; the overview, ch. 6.

- **What rests on calculations.** No calculation: it is an argument. It concerns the laws; single values, which MM calls history, come from measurement.
- **Open.** A different law measured in a distant region would take away this reading's support.

3.3 Identical particles (way A)

Proposal of the AI, not contained in earlier deposits; the author accepted it on 28 October 2026.

- **What is observed.** All electrons have the same mass and the same charge, and they are indistinguishable in the strict sense: exchanging two of them changes no prediction. For fermions the state changes sign under exchange; which sign holds for which species is fixed by the spin-statistics theorem.
- **Known physics.** In field theory every electron is an excitation of the same field; why there is one field per species is not asked. Feynman recounts, in his Nobel lecture (1965), that Wheeler told him on the telephone that all electrons have the same charge and the same mass «because they are all the same electron». In philosophy it is debated whether two identical particles are two objects; Saunders (2003) argues that they are, in a weak sense.
- **What the non-local phase does.** Two electrons are two copies of the same unit, counted twice: they are indistinguishable because between the copies there is nothing that makes them different. The reading is close to that of field theory: the common field is the common nexus.
- **Where the link comes from.** A proposal of the AI.
- **What rests on calculations.** No calculation. The step from copies to the sign of exchange has not been made.
- **Open.** Why different copies give different species, such as the electron and the muon: without this step the argument would prove too much.

3.4 The objects that two observers see alike (ways A and B)

- **What is observed.** Two people look at the same chair with different photons and agree on where it is. In laboratories a piece of information becomes objective when many copies of it spread into the environment, and whoever reads a small part of it finds the same value (quantum Darwinism, Zurek 2009).
- **Known physics.** Quantum Darwinism explains agreement with the redundancy of copies. The Wigner's-friend theorems (Bong et al. 2020) show that, if quantum mechanics holds also for an observer controllable as a quantum system, the absoluteness of outcomes, locality and the absence of superdeterminism cannot all hold together.
- **What the non-local phase does.** The two records are two counts of the same non-local relation, made from two different points; their common part is the trace of that relation (way A). The observed object arises as a non-local transposition and projection of the observer's matrix: by projecting its own matrix and discarding the private part, the observer isolates the trace of the relation, which has no distance between its copies. It is the same idea with which the framework reads entanglement: the two distant measurements exchange nothing; they are the two ends of a single relation. And there is a prohibition: two observers who have enough of the object cannot disagree. The hardest side: what has no non-local link with us casts no shadow in our record, and for us it does not exist (way B). For a laboratory that lets no copies out, MM gives up the absoluteness of the outcome (2.48, §5).
- **Where the link comes from.** From the deposits (2.62.1, §2.2 and §2.5) and from the author's premises of 12, 13 and 23 October.
- **What rests on calculations (MM 2.62 and 2.62.1).** In a quantum-Darwinism model the common object is the Gács–Körner common part of the two records. With tolerance 10^{-6}, from 23 copies per observer the common part is almost the whole object: about 1 bit, almost all of the question «where is the chair?». The exact common part, without tolerance, is zero until the records are compared. With an object with more values the common part grows in steps.
- **Open.** The calculations test the common part, not the non-local phase. Identifying the copies of the framework with scattered photons is a choice of the AI. Agreement between readings is already guaranteed by standard quantum mechanics: MM recovers it; it does not predict it in its own right.

3.5 The boundary of the local is a count (ways A and C)

- **What is observed.** In an atom interferometer, a single scattered photon erases the fringes when the two paths are about half a wavelength apart, with revivals beyond (Kokorowski et al. 2001). Heated fullerenes lose their fringes (47 %, 29 %, 7 %, 0 %) through thermal photons that fly away (Hackermüller et al. 2004). The copies carried away count even if nobody collects them.
- **Known physics.** This is decoherence: the passage to the classical depends on how many copies the environment carries away and on the separation compared with the wavelength, not on the distance from whoever records.
- **What the non-local phase does.** If the local is the non-local counted, its boundary cannot be a distance from the observer: it is a number of copies. An ordinary record, 1 cm^2 at 300 K, leaves about 4×10^{12} copies of itself in one microsecond; a few tens suffice for a complete reading (21 for 99 % of the information, with copies that

overlap by 0.9). At zero copies no local reading exists.

- **Where the link comes from.** From the status report 2.9-2.18 (QUANTUM_PLACEMENT, §0), reread here in the pre-geometric sense. That August text used «non-local» in Bell's sense; 2.48 corrected it.
- **What rests on calculations.** The count of copies (plateau_copie.py, status report 2.9-2.18; the same number, 21 copies for a 1 % deficit, in 2.62.1); reading D1 of 2.66, compatible with standard decoherence.
- **Open.** The step «matrix → place», that is, which correlations follow from the count, has not been made. Agreement between readings with enough copies is guaranteed by standard quantum mechanics; MM recovers it.

3.6 Probability (way C)

- **What is observed.** A particle sent alone towards two slits arrives at a single point; many particles, one after another, draw the fringes of a wave. The probability of each point is the square of an amplitude: the Born rule.
- **Known physics.** The Born rule is a postulate. Derivations from counting branches or copies (Zurek 2005) rest on a contested step: equal copies, equal probabilities (Schlosshauer and Fine 2005).
- **What the non-local phase does.** For the author the non-local phase and the mirror reciprocity between the two ends of the nexus give a reason for the probabilistic behaviour of matter. The framework's image: having to describe a 10 with the number 1 alone forces one to distribute the 1 over ten positions, one for each unit of the 10.
- **Where the link comes from.** From the author's premise of 22 October, in the summary «MM in brief» on all records.
- **What rests on calculations (MM 2.40, 2.48 and 2.48.1).** Counting copies of equal modulus gives $|\psi|^2$ with three declared assumptions, in the cases tested (ratios with denominator up to 12), plus continuity for irrational ratios; among the exponents tested, the step of the 2.40 checkerboard conserves only the sum of squares. In the loop the Born weight is contained in the closure rule: for the Born rule the loop reading is circular, and the measure over futures remains a postulate. A self-consistency argument (2.48, R3), in the family of protocols tested (pair model, angular law $\cos^2$ assumed), shows that the weight of consistent histories with weights other than Born's becomes as small as one likes as the number of pairs grows, but stays positive for every finite number: impossibility is not shown, and the step to «they do not exist» is a postulate of the author.
- **Open.** That probabilities exist, and which outcome occurs, are not derived.

3.7 When an object stops being at one point (way C)

- **What is observed.** Today nanoparticles of more than 7 000 atoms and 170 000 atomic mass units interfere (Pedalino et al. 2026), while for everyday objects, under ordinary conditions, no interference between distinct positions is observed. No experiment has yet found the boundary.
- **Known physics.** Decoherence explains the classical look of large bodies; collapse models, such as that of Diósi and Penrose, look for a true limit, present even in a vacuum.
- **What the non-local phase does.** When the object is large enough, its mass fixes it in the matrix of the point of observation: we see it in one place and at one instant. Below a certain level of mass, the GCD of the point of observation and that of the diffraction that generates the object become equal, as in a mirror: the object no longer occupies a single point of the observer's spacetime and spreads over several positions. In the framework's reading the boundary is the point at which the object becomes the common part of the records (the bridge with §3.4).
- **Where the link comes from.** From the author's premise of 22 October («the logical mirror system and the non-local phase account for the probabilistic behaviour of matter») and from 2.66.
- **What rests on calculations (MM 2.66).** Ten readings of the threshold as a mass and eight of the distance, registered before the calculations. The two readings of distance that hold, one for the state (D1) and one for the reading (D6g), coincide with decoherence and with wave optics: the framework admits them; it does not predict them differently. The reading with the 2.40 clock describes a condition of the apparatus, that is, of the reading, not a property of the object. A prediction dated 31 December 2035 is registered with a program that anyone can apply; the most likely outcome in 2035 is «not assessable», because the planned experiments do not reach the effect.
- **Open.** The number for the threshold; whether D1 and D6g are a single rule of the framework (a proposal of 2.66).

3.8 The present and the choice of what is written (way C)

- **What is observed.** Every measurement gives a single outcome, and what is written stays written. In some experiments, with initial correlations, heat flows from cold to hot (Micadei et al. 2019).
- **Known physics.** The present has no place in the equations; the arrow of time is tied to heat and to initial conditions.
- **What the non-local phase does.** For the author the non-local basis forces the so-called present to a part of the

matrix to which the observer belongs: the non-local link «switches on» only part of the total matrix. Also for the author, which reading is written is decided by the future of the history, which chooses its past: the loop. The non-local phase has no before or after; in the AI's reading the loop is how that unity appears in the local reading.

- **Where the link comes from.** From the author's premises of 11 and 13 October, published in 2.48 and 2.68; the translation of the loop into a model in light cones is the AI's (2.48).
- **What rests on calculations (MM 2.48, 2.64.1 and 2.68.1).** In the single calculation of 2.68.1, loop, record and direction of time stand in one model. With the rule in which the future chooses by the closure of the loop, the first readings have the same statistics whether the history stops or continues, the record grows and gives the Born frequencies, and in the Micadei testbed it grows while heat flows backwards, on average and along the most probable history. On the 2.40 checkerboard the record grows less than the threshold set beforehand (0.29 and 0.17 bits against 0.5), and «only the past» does not by itself select the Born rule.
- **Open.** What distinguishes the «switched-on» part of the matrix from the part that is switched off (2.68, §10).

3.9 Space and time themselves (way A)

- **What is observed.** Moving clocks run slow, as relativity requires: proper time is a fact measured with atomic clocks.
- **Known physics.** Spacetime is the starting stage. In theories of quantum gravity one looks instead for a spacetime that emerges from something that is not spacetime (Huggett and Wüthrich 2025); for some, the connectedness of space comes from entanglement (Van Raamsdonk 2010).
- **What the non-local phase does.** For the author a non-local layer has in itself the features to generate space and objects. If the local is the non-local counted, places are not given beforehand: they arise with the count. A clock that counts events without pre-assigned places must give the time we measure, and places given beforehand must act as a privileged frame.
- **Where the link comes from.** From the author's premise of 10 October; the link with the calculations of 2.40 is the AI's, because 2.40 speaks of events without places, not of the non-local phase.
- **What rests on calculations (MM 2.40 and 2.40.2).** A clock that reads the number of copies runs at the geometric mean of the two light sides, that is, at Einstein's proper time; given the premise of counts on the two sides (as in Bondi), the factor $\sqrt{(1 - v^2)}$ is an identity. Without pre-assigned places, with copies at random, the longest chain recovers it within about 1 % in 1+1 dimensions; on a grid of places the same reading gives the arithmetic mean, Galilean time; with fixed places and random times the dilation is 7 to 29 % stronger than Lorentz's, because the places define a frame. In 3+1 dimensions, with about 6 000 copies, the comparison between clocks with the same proper time holds within 1.3 σ; at $v = 0.8$, with about 780 copies, the deviation exceeds the fixed 6 % threshold registered for the P1-P5 block. With more than two counted sides only the pairwise rule keeps the Lorentz factor, and n sides give $n - 1$ directions. Neighbours: Knuth and Bahreyni (2014).
- **Open.** The number of dimensions is not derived; gravity, Lorentz invariance and 3+1 are not derived from the pre-geometric programme alone (2.30).

3.10 The limit of light (way A)

- **What is observed.** No body and no signal locally exceeds the speed of light in vacuum, which is the same for all observers.
- **Known physics.** In relativity it is a starting principle; since 1983 the metre has been defined from it.
- **What the non-local phase does.** The speed of light is the fixed rate of exchange between the multiples and the fractions of the same GCD: how much space one step of time is worth. Exceeding it would mean, for some observer, covering space without counting time, that is, leaving the local world. In the AI's reading the rate is the same for everyone because every observer counts copies of the same nexus, as for the laws of §3.2. Outside the system there is no other term to refer it to.
- **Where the link comes from.** The fixed rate, and exceeding it as leaving the local world, are in the summary «MM in brief», approved by the author, and in 2.68, §7; the link with copies of the same nexus is the AI's.
- **What rests on calculations.** No calculation. The closest calculation is the geometric mean of 2.40 (§3.9).
- **Open.** Why the rate has precisely the measured value: inside the system it is a necessary relative; outside, it has no justification.

3.11 Dark matter as a second mirror (way B)

- **What is observed.** Galaxies and clusters move as if there were about five times more matter than the matter that emits or absorbs light; in the standard cosmological model, Planck's data give a cold non-baryonic component about 5.4 times the baryon density (Planck Collaboration 2020).
- **Known physics.** One proposed explanation is a new particle, which gravitates and does not interact with light; no candidate has been confirmed. There are also models in which dark matter is ordinary matter of a mirror sector (Kolb, Seckel and Turner 1985; Berezhiani 2004; Foot 2004).

- **What the non-local phase does.** For the author dark matter is ordinary matter of another mirror. A large part of it is real for its own local GCD, to which it is bound non-locally; not being bound non-locally to the observer's GCD, in the principal mirror it is non-existent. There is a reciprocal non-existence for lack of a reciprocal non-local link. Yet it gravitates: for the author the fractionation acts on the electromagnetic interaction, not on the gravitational one. In way B, then, «it does not exist for us» holds for light and for electromagnetic interactions, not for gravity. For the author, finally, what decides the difference between the two mirrors is the non-local share of exchange within each.
- **Where the link comes from.** From the author's premises reported in 2.44 (words 2, 5, 8 and 12).
- **What rests on calculations (MM 2.44).** A hidden sector with its own light, and the numerical hypothesis $\Omega_c/\Omega_b = 3\sqrt{3}$, supplied separately by MM. Imposing it costs $T_B = 0.17$-0.23 with Planck and DESI DR2: compatible, in a retrospective check; a test on future data is registered. The words on the share of exchange motivated the form of the radial acceleration relation (RAR) written in terms of L/S, which is not a derivation. The reading also postulates that hidden radiation does not source our expansion, and the worked example does not establish a viable hidden sector.
- **Open.** The relative constants of the second mirror are left in a generic form; a complete calculation of the hidden sector is missing.

3.12 The boundary of a gravitating system, set by the Galaxy (way C)

- **What is observed.** A wide binary star near the Sun sits in the field of the Galaxy. At low accelerations galaxies follow a common scale, a_0, and in MOND the internal dynamics of a system also depends on the external field it sits in.
- **Known physics.** In Einstein's gravity the strong equivalence principle holds: a uniform external field does not change the internal dynamics of a system. In MOND this principle does not hold.
- **What the non-local phase does.** What a system reads depends on the context it belongs to, not on the point. For the author the behaviour depends on the observational context, and the Galaxy acts on the observation GCD, which in turn determines the system's own field. In the AI's reading, here the non-local shows as context: the law of a system depends on the whole, as in a reading and not as in a force.
- **Where the link comes from.** From the author's premises of 27 and 28 October. In 2.70 the hypothesis «a non-local element of context is needed» is said to agree with the premises of 10 October (the non-local layer) and of 27 October: consistency, not proof.
- **What rests on calculations (MM 2.70).** The local versions of the junction between MM's dynamics and the a_0 scale tried so far do not pass the comparison with Cassini (13.7 to 23.5 σ beyond the limit) or the granularity test; the hypothesis of a non-local context remains motivated, not verified, and the tests do not show that it is necessary. Translated by the AI as «the boost with which a system is read does not exceed the boost with which the Galaxy reads the point» (T_{max}), the premise of 28 October gives back the boundary, where the system's own field falls to the level of the external one, and the isotropy inside the boundary, which before were only posited. On the Gaia DR3 pairs T_{max} is the best of eight hypotheses, but a posteriori, and it does not separate from MM's chain. For the globular cluster NGC 2419 the law gives a boost of 3.3-3.4 at 86 pc, against $\lesssim 1.1$ from the data, like every law made of the internal and external accelerations alone; the proposed way out is the history of the cluster, which the premise does not replace. The rule for Gaia DR4 is registered.
- **Open.** The translation has two declared choices (scalar and maximum), and why the maximum is not derived. With DR4 the most likely outcome is «not decidable», both for the amplitude and for the direction: with the direction observable (Part A of 2.70) about ten times the DR3 pairs would be needed to reach 2σ.

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4. Summary table

Origin of the link with the non-local phase: **D** already written in a deposit; **A** premise of the author; **AI** proposed by the AI. The column says where the link between the non-local phase and the phenomenon comes from; the translations into models and calculations are always the AI's and are stated under the calculations (for example §3.4, §3.8, §3.12).

phenomenon	way	what the non-local phase does	origin	what rests on calculations	deposits
correlations of entanglement	A, C	the two ends are one nexus; the ceiling is set by the reading	D	Tsirelson with local Born; loop for the singlet	2.48, 2.48.1
same laws everywhere	A	every here and every now is a copy of the same nexus	D	argument	2.68, 2.68.1
identical particles	A	two electrons are the same unit	AI	none	—

		counted twice			
shared objects	A, B	two records count the same relation; without a link nothing exists for us	D, A	common part in a quantum-Darwinism model	2.62, 2.62.1
boundary of the local	A, C	the boundary is a number of copies	D (reread)	count of copies; decoherence	status report 2.9-2.18, 2.62.1, 2.66
probability	C	describing 10 with 1 forces one to distribute it	A	Born from counting with three assumptions	2.40, 2.48, 2.48.1
quantum threshold	C	below a mass the reading that fixes gives way	A, D	registered readings; prediction for 2035	2.66
present and writing of outcomes	C	the non-local link switches on part of the matrix; the loop chooses	A, D	single calculation of loop, record and direction	2.48, 2.64.1, 2.68.1
space and time	A	places arise with the count	A	proper time from events without places	2.30, 2.40, 2.40.2
limit of light	A	the same rate for whoever counts copies of the same nexus	D, AI	none	«MM in brief», 2.68
dark matter	B	without a reciprocal non-local link there is no common light	A	hidden sector; 3√3 imposed, compatible a posteriori	2.44
boundary of gravitating systems	C	the Galaxy fixes the reading of the system	A, D	local versions beyond the Cassini or granularity limits; T_max on DR3 a posteriori; DR4 test registered	2.70

In twelve phenomena far apart, from the scale of photons to that of the Galaxy, MM's reading uses the same premise in the same way. In ten the link is already in the deposits or in the author's premises; in one it comes partly from the AI, and in one from the AI alone. This unity of the reading is the result of the deposit: in MM's reading non-locality is not an exception of entanglement, but the common ground of facts that known physics explains with separate assumptions.

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5. What is not claimed

- The non-local phase transmits no signals and no influences outside light cones.
- Today it has no distinct experimental signature. Where MM gives numbers, they coincide with those of known physics (the Tsirelson bound, decoherence, wave optics) or come from readings with declared choices (T_max, a posteriori).
- The readings are not the only possible ones: the same observations have other interpretations, and experiments do not choose among them.
- The deposit does not solve the measurement problem and does not derive the Born rule without assumptions.
- The translations of the premises into calculations are the AI's and are declared as such in the deposits cited.

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6. Where the reading is put to the test

- **Gaia DR4 (December 2026).** Rules registered in 2.70 for the boundary set by the Galaxy (§3.12). The date fixes the test; the most likely outcome is «not decidable».
- **The mass threshold (31 December 2035).** The dated prediction of 2.66, applicable by anyone with the deposited program (§3.7).
- **Experiments that change only the distance from whoever records.** There are none today (2.66, §4.6); they would make the role of distance testable (§3.7).
- **Five open questions** (2.68, §10):
- if in the non-local phase there is no measure, no before and no after, what makes the original GCD begin to be counted;
- what distinguishes the «switched-on» part of the matrix from the part that is switched off;
- in what sense the reading of a record is non-local;

- in entanglement, what relation there is between the non-local phase and the loop;
- if the non-local phase is a real fact, what is the main physical consequence that today's physics sees but does not explain.

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7. One word, several senses: «non-local» elsewhere in the deposits

In MM's records the word also appears in different senses, which must not be confused with the non-local phase.

- **Covariant memory** (August status report, §7): a factor that depends on the integral of the curvature over the past light cone, from the family of non-local gravity cited in that text, and a memory along the trajectory of the body. This is non-locality in the sense of field theory. The August text itself says that the two uses coincide «by linguistic accident».
- **The horizon coupling** $r_t = \sqrt{(R_s \cdot R_H)}$ (status report 2.9): a local scale tied to the cosmic horizon. The $a_0(z)$ adopted later uses the elapsed-age clock, which is local.
- **Matter in compact stars** (2.34.2): through ψ the stress at a point depends on the whole configuration.
- **The reading operator on the lattice** (2.52): locality in the sense of lattice fermions.
- **The law L*** (2.70): «non-local» because it depends on the whole system and not on the point. It is tied to the non-local phase only through the author's premises, as consistency (§3.12).

The «Bell» sense used in the August texts on the quantum placement (status report 2.7-2.27) was also corrected in 2.48: in MM «non-local» means not located in space and time.

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8. Neighbours

- **The implicate order** of Bohm (1980): what appears separate is enfolded in an undivided whole.
- **Non-separability** as a thesis distinct from locality: Howard (1985), who reads it in Einstein; Teller (1986), with relational holism.
- **Emergent spacetime:** Huggett and Wüthrich (2025); the connectedness of space from entanglement, Van Raamsdonk (2010); spacetime from an influence network, Knuth and Bahreyni (2014).
- **Bell non-locality and relativity:** Maudlin (2011).
- **Retrocausal readings with local mediation:** Aharonov, Bergmann and Lebowitz (1964); Cramer (1986); Argaman (2010); Wharton and Argaman (2020).
- **Objectivity from copies:** Zurek (2009); relational quantum mechanics, Rovelli (1996).
- **Identical particles:** Feynman (1965) on Wheeler's idea; Saunders (2003) on weak discernibility.
- **Dark matter as matter of a mirror sector:** Kolb, Seckel and Turner (1985); Berezhiani (2004); Foot (2004).

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9. Who did what, and how this text was made

The premise of the non-local phase and the sentences from which the readings start are the author's, Ugo Lanciano, who is not a physicist; his exact words, with dates, are in AUTHOR_BACKGROUND.md. The translations of the premises into calculations, the calculations and the texts of the deposits cited were made with the AI (Claude), with checks by other systems (Astra, Gemini) described in each deposit.

This text was written by the AI in Italian on 28 October 2026, at the author's request, from the published records, read in local copies identical to the deposited files; the English text is a translation of the Italian one. A second instance of the same AI compared the first draft with the texts of the deposits and made 21 remarks, all checked against the sources and applied (REVIEWS/REVIEW_1_CLAUDE_IT.md). The author read the second draft and accepted it, including the two proposals of the AI (§2 and §3.3). Then two external systems, Astra (Codex) and Gemini (Argon), checked the final texts in two rounds against extracts of the deposits cited: 17 remarks in the first round, almost all applied, and one rejected on the merits with the reason; in the second round Gemini answered «ready» and Astra «ready with corrections», with three remarks applied (REVIEWS/REVIEW_2_ASTRA_ARGON_G1_IT.md and REVIEWS/REVIEW_3_ASTRA_ARGON_G2_IT.md). The new references were checked on open pages (REFERENCE_CHECK.md).

Glossary

- **Non-local phase:** reality as pure relations, without space, without time and without objects; present everywhere, neither before nor beneath the rest.
- **Local phase, matrix:** the same reality counted from a point; the multiples of the GCD appear as time, the fractions as space.
- **Nexus:** the relation between two distinct terms that have a single element in common, as 3 and 5.
- **GCD:** the common element of the two terms of a nexus (in Italian MCD).
- **Copy:** the same unit counted once more; between two copies there is no distance.
- **Reading:** what an observer, that is, any part of the structure that records and compares, obtains from its own matrix.
- **Loop:** the closure of a history between preparation and outcome, in which the future contributes to choosing the past.

Deposits cited

record	what	DOI
status report 2.7-2.27	the August quantum placement, the count of copies	10.5281/zenodo.21802475 (concept)
entry page 2.27	the non-local phase present everywhere	10.5281/zenodo.22667516
2.30	the pre-geometric programme	10.5281/zenodo.22730712
2.40, 2.40.1, 2.40.2	clock without places, checkerboard, pairwise rule	10.5281/zenodo.23280167 (concept); 2.40.2: 10.5281/zenodo.23631239
2.44	dark matter as a second mirror	10.5281/zenodo.23329080
2.48	Tsirelson bound, loop	10.5281/zenodo.23352814
2.48.1	Born from counting copies	10.5281/zenodo.23655646
2.62, 2.62.1	objects as common part	10.5281/zenodo.23609966 (concept); 2.62.1: 10.5281/zenodo.23643717
2.64.1	record and direction of time	10.5281/zenodo.23643718
2.66	mass threshold, distance	10.5281/zenodo.23643759
2.68, 2.68.1	what «non-local» means, single calculation	10.5281/zenodo.23669889 (concept); 2.68.1: 10.5281/zenodo.23672394
2.70	boundary set by the Galaxy, DR4 test	10.5281/zenodo.23730218
overview	the framework as a whole	10.5281/zenodo.23535916 (concept)

References

Each reference is linked to the open page on which it was checked: in deposits 2.44, 2.48, 2.62.1, 2.66 and 2.68 for those already used there, and on 28 October 2026 for the new ones (REFERENCE_CHECK.md).

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Contents of the package

file	what
README.md	this text
LEGGIMI_IT.md	the same text in Italian (the original)
AUTHOR_BACKGROUND.md	the author's sentences on the non-local phase, with exact words and dates
REFERENCE_CHECK.md	how the new references were checked
REVIEWS/	the reports of the checks on the text
LICENSE.md	licence (CC BY 4.0)
IMPRONTE.sha256	SHA-256 fingerprints of the files

There is no code in this deposit: the calculations it cites are in the deposits listed above.